

Teaching Exam

Subject – Computer Science

**Topic – Computer Organization &
Architecture**



Lecture No. – 07

By – Rahul Sir

Topics to be Covered

Topic



- 1:- Decoder and its Various Types ✓
- 2:- Application of Decoder ✓
- 3:- Comparator ✓
- 4:- Types Of Comparator ✓
- 5:- 1,2,4-bit Comparator ✓
- 6:- IC 7485 ✓
- 7:- Application and Conclusion of Comparator ✓



In digital electronics, a combinational logic circuit that converts an N-bit binary input code into M output channels in such a way that only one output channel is activated for each one of the possible combinations of inputs is known as a **decoder**.

In other words, a combinational logic circuit which converts N input lines into a maximum of 2^N output lines is called a **decoder**.

Therefore, a decoder is a combination logic circuit that is capable of identifying or detecting a particular code. The operation that a decoder performs is referred to as decoding. A general block diagram of a decoder is shown in Figure-1.

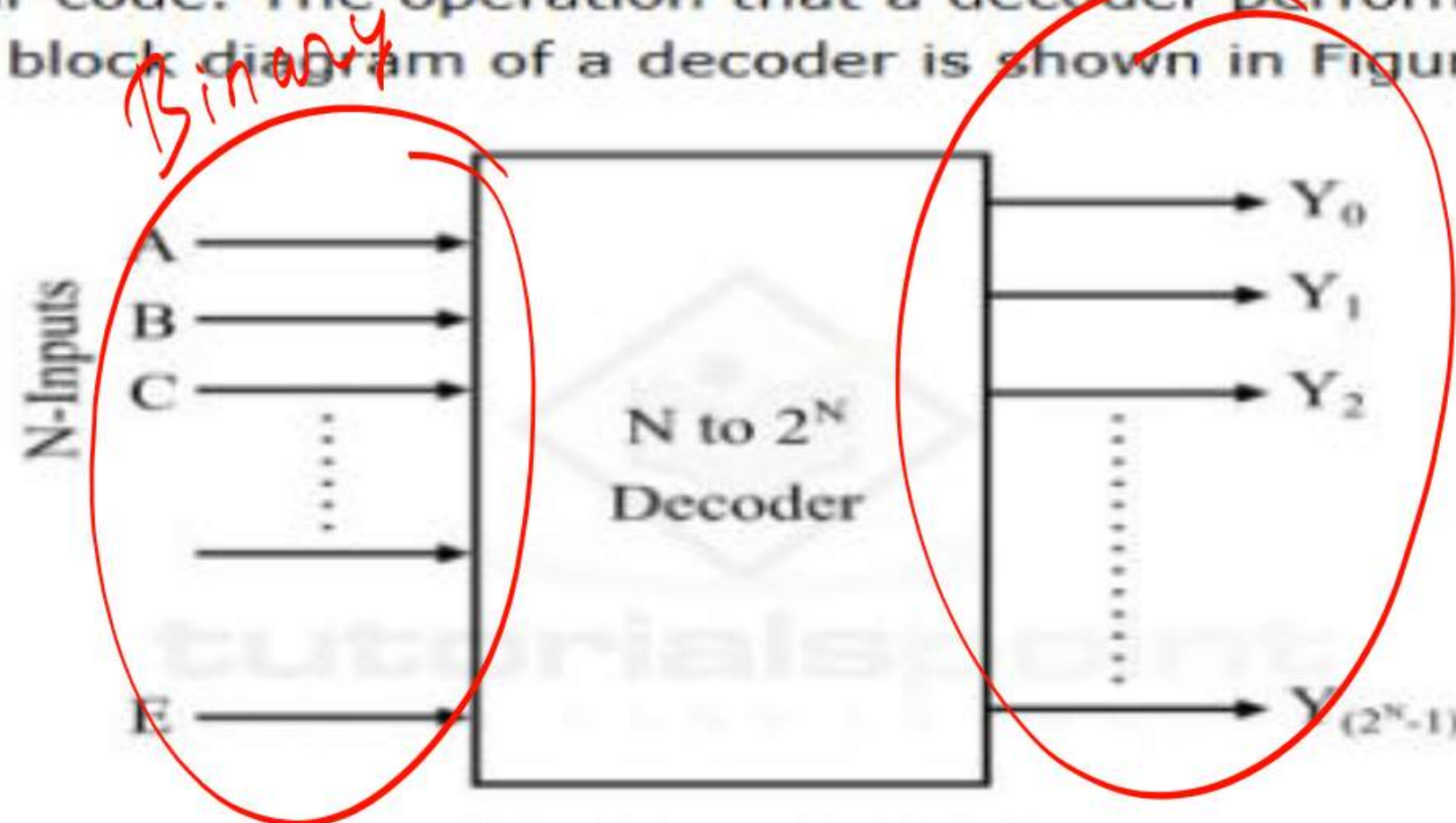


Figure 1 - Decoder

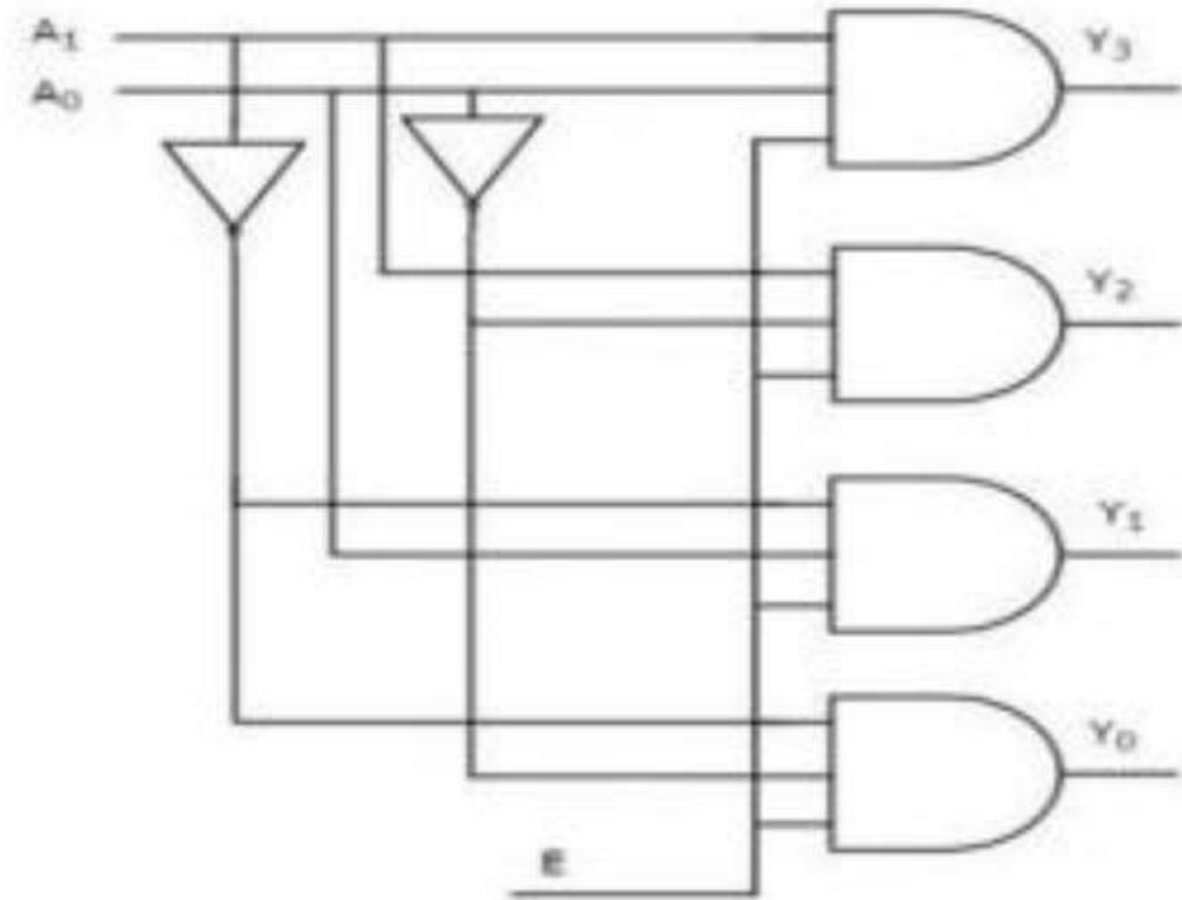
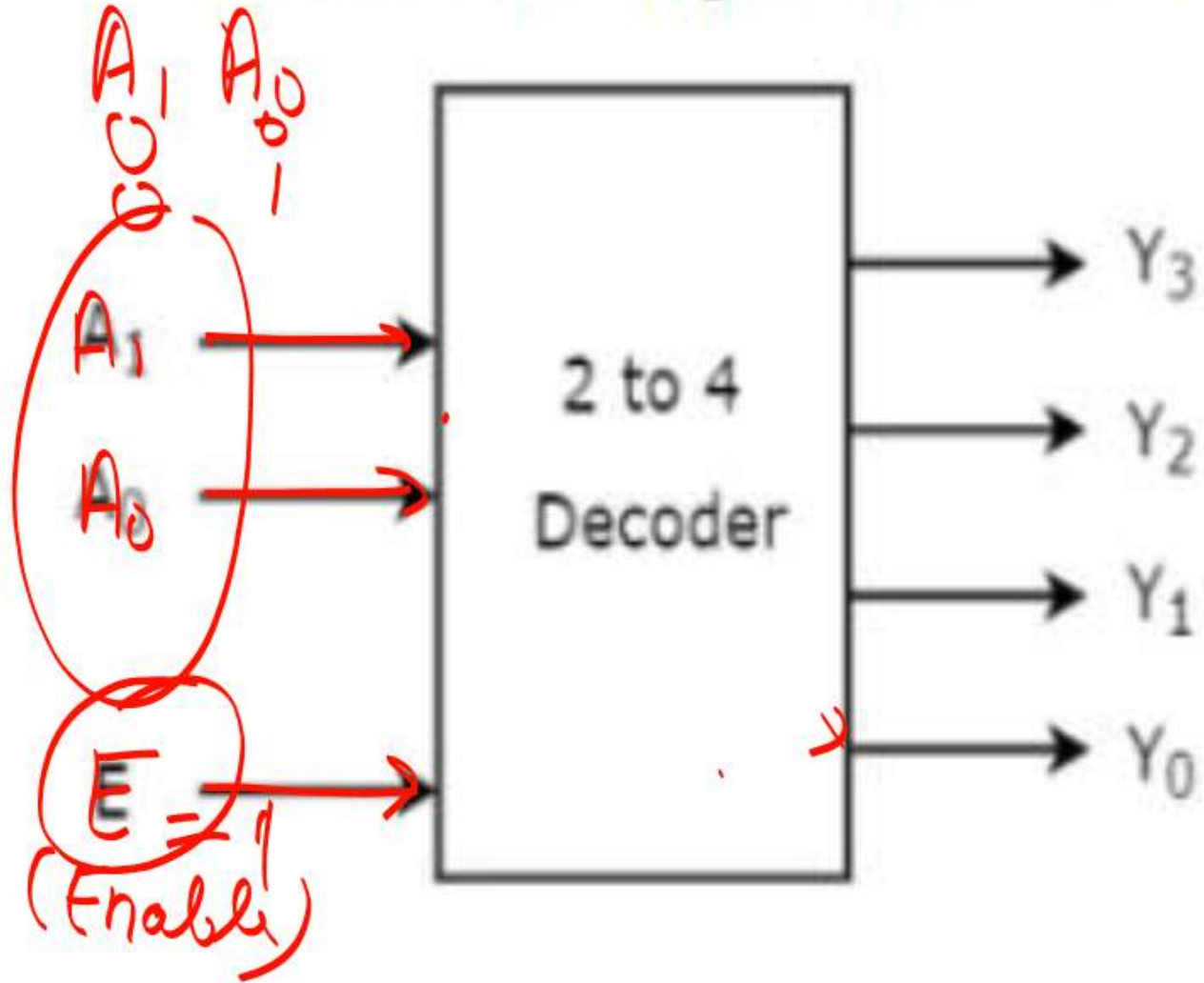


Here, the decoder has N input lines and M (2^N) output lines. In a decoder, each of the N input lines can be a 0 or a 1, hence the number of possible input combinations or codes be equal to 2^N . For each of these input combinations, only one of the M output lines will be active, and all other output lines will remain inactive.

**Note:- Binary Bit at input are Converted in Octal,
Decimal and Hexadecimal at output in Decoder**



The 2 to 4 decoder is one that has 2 input lines and 4 (2^2) output lines. The functional block diagram of the 2 to 4 decoder is shown in Figure-2.



When this decoder is enabled with the help of enable input E , then its one of the four outputs will be active for each combination of inputs. The operation of this 2-line to 4-line decoder can be analyzed with the help of its truth table which is given below.



Topic : 2 to 4 Decoder



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Inputs			Outputs			
E	A ₁	A ₀	Y ₃	Y ₂	Y ₁	Y ₀
0	X	X	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

Using this truth table, we can derive the Boolean expression for each output as follows –

$$E = 1$$
$$\overline{E} = 0$$

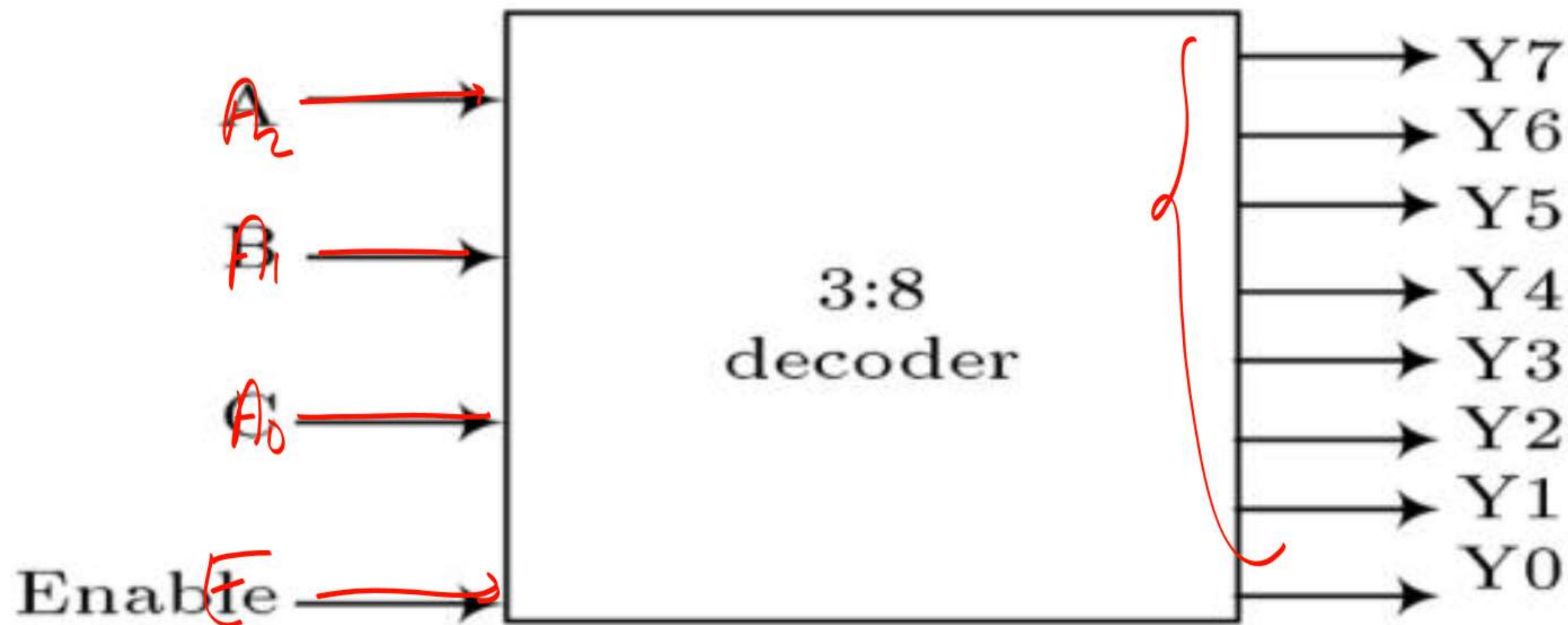
$$A_1 = 1, A_0 = 1$$
$$\overline{A_1} = 0, \overline{A_0} = 0$$

$$Y_0 = E \cdot \overline{A_1} \cdot \overline{A_0}$$
$$Y_1 = E \cdot \overline{A_1} \cdot A_0$$
$$Y_2 = E \cdot A_1 \cdot \overline{A_0}$$
$$Y_3 = E \cdot A_1 \cdot A_0$$



3 to 8 Decoder

The 3 to 8 decoder is one that has 3 input lines and 8 (2^3) output lines. The functional block diagram of the 3 to 8 decoder is shown in Figure-4.



When this decoder is enabled with the help of enable input E, then it's one of the eight outputs will be active for each combination of inputs. The operation of this 3-line to 8-line decoder can be analyzed with the help of its function table which is given below.



Topic : 3 to 8 Decoder



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Power check

Binary Input

Inputs				Outputs							
E	A ₂	A ₁	A ₀	Y ₇	Y ₆	Y ₅	Y ₄	Y ₃	Y ₂	Y ₁	Y ₀
0	x	x	x	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	1
1	0	0	1	0	0	0	0	0	0	1	0
1	0	1	0	0	0	0	0	0	1	0	0
1	0	1	1	0	0	0	0	1	0	0	0
1	1	0	0	0	0	0	1	0	0	0	0
1	1	0	1	0	0	1	0	0	0	0	0
1	1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	1	0	0	0	0	0	0	0

$Y_0 = E \bar{A} \bar{B} \bar{C}$

$Y_1 = E \bar{A} \bar{B} C$

$Y_2 = E \bar{A} B \bar{C}$

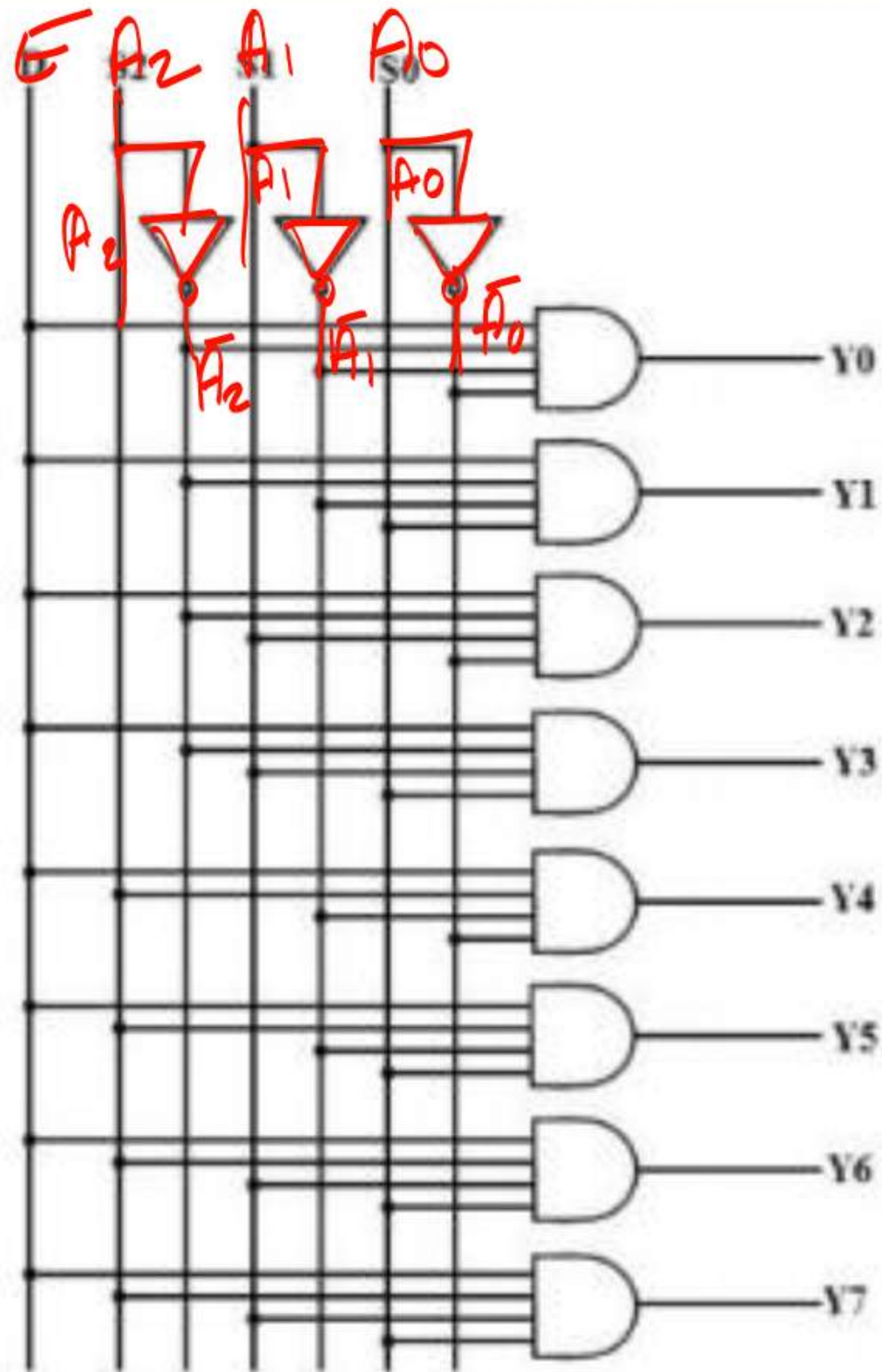
$Y_3 = E \bar{A} B C$

$Y_4 = E A \bar{B} \bar{C}$

$Y_5 = E A \bar{B} C$

$Y_6 = E A B \bar{C}$

$Y_7 = E A B C$



$$Y_0 = E \bar{A} \bar{B} \bar{C}$$

$$Y_1 = E \bar{A} \bar{B} C$$

$$Y_2 = E \bar{A} B \bar{C}$$

$$Y_3 = E \bar{A} B C$$

$$Y_4 = E A \bar{B} \bar{C}$$

$$Y_5 = E A \bar{B} C$$

$$Y_6 = E A B \bar{C}$$

$$Y_7 = E A B C$$



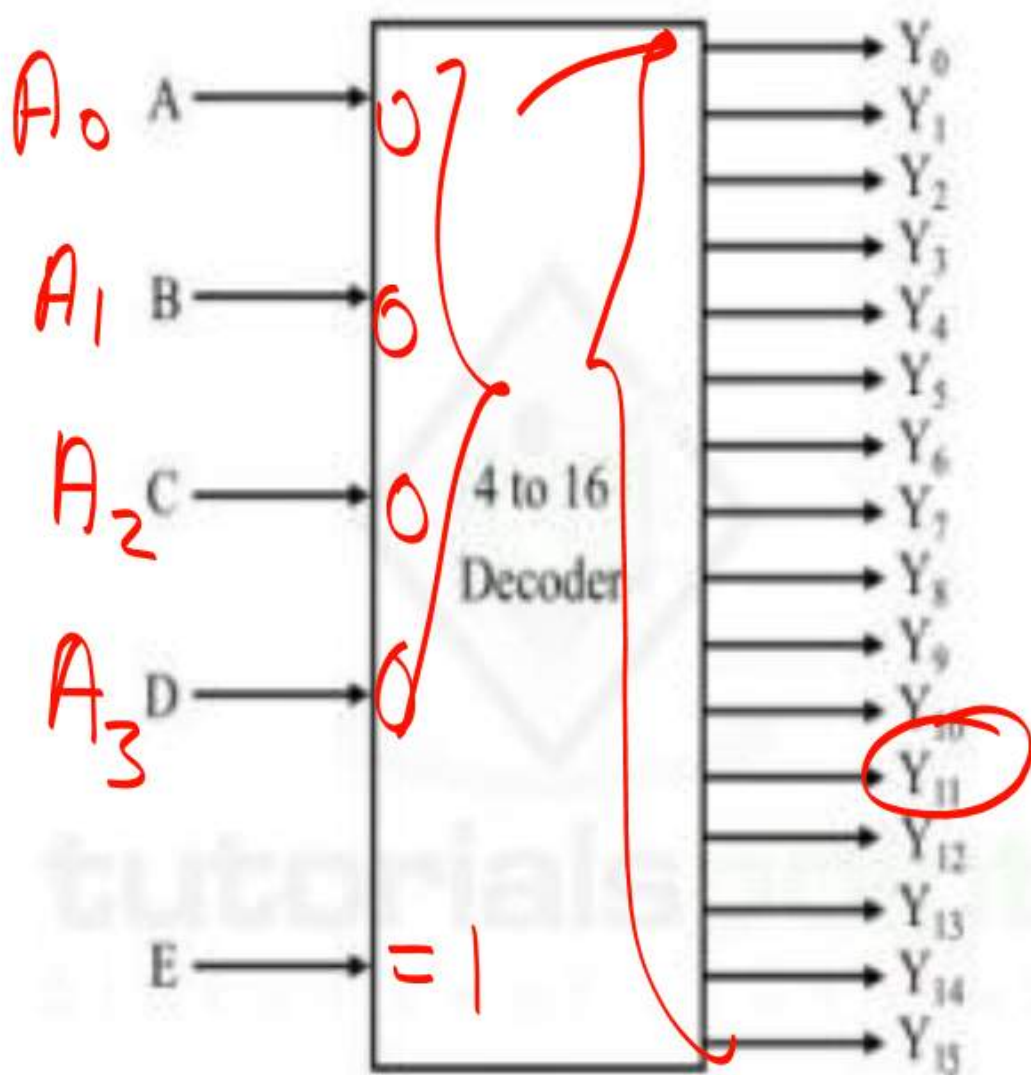
Topic : 4 to 16 Decoder



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4 to 16 Decoder

The 4 to 16 decoder is the type of decoder which has 4 input lines and 16 (2^4) output lines. The functional block diagram of the 4 to 16 decoder is shown in Figure-6.



$$E A_3 A_2 A_1 A_0$$

$$Y_0 = E \bar{A} \bar{B} \bar{C} \bar{D}$$

$$Y_1 = E \bar{A} \bar{B} \bar{C} D$$

$$Y_2 = E \bar{A} \bar{B} C \bar{D}$$

$$Y_3 = E \bar{A} \bar{B} C D$$

$$Y_4 = E \bar{A} B \bar{C} \bar{D}$$

$$Y_5 = E \bar{A} B \bar{C} D$$

$$Y_6 = E \bar{A} B C \bar{D}$$

$$Y_7 = E \bar{A} B C D$$

$$Y_8 = E A \bar{B} \bar{C} \bar{D}$$

$$Y_9 = E A \bar{B} \bar{C} D$$

$$Y_{10} = E A \bar{B} C \bar{D}$$

$$Y_{11} = E A \bar{B} C D$$

$$Y_{12} = E A B \bar{C} \bar{D}$$

$$Y_{13} = E A B \bar{C} D$$

$$Y_{11} = E A_3 A_2^1 A_1 A_0$$

$$Y_{14} = E A B C \bar{D}$$

$$Y_{15} = E A B C D$$



Topic : 4 to 16 Decoder



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Inputs					Output
E	A	B	C	D	
0	X	X	X	X	0
1	0	0	0	0	Y_0
1	0	0	0	1	Y_1
1	0	0	1	0	Y_2
1	0	0	1	1	Y_3
1	0	1	0	0	Y_4
1	0	1	0	1	Y_5
1	0	1	1	0	Y_6
1	0	1	1	1	Y_7
1	1	0	0	0	Y_8

1	1	0	0	1	Y_9
1	1	0	1	0	Y_{10}
1	1	0	1	1	Y_{11}
1	1	1	0	0	Y_{12}
1	1	1	0	1	Y_{13}
1	1	1	1	0	Y_{14}
1	1	1	1	1	Y_{15}



Applications of Decoders

Decoders are used in the cases where an output or a collection of outputs is to be activated only on the occurrence of a particular combination of input codes. Some important applications of decoders are listed below –

- Decoders are used for code conversions. ✓
- Decoders are extensively used in memory systems of computers. ✓
- Decoders are also used for de-multiplexing or data distribution. ✓
- Decoders are also used in data routing applications where very short propagation delay is required.
- Decoder may also be used for timing or sequencing purposes.
- Decoders are also utilized to turn on and off digital devices at a specific time.





In digital electronics, a **comparator** is a combinational logic circuit that is used to compare the magnitudes of two binary numbers. Comparators are used in several different electronic circuits like analog to digital converters, voltage level detectors, zero-crossing detectors, etc. The most basic example of a comparator is an XNOR gate which produces a high or logic 1 output only when both its inputs are equal.

What is a Comparator?

A digital combinational circuit used to compare the magnitude of two binary numbers to determine the equality or non-equality is called a comparator.

Therefore, the main function of a comparator is to compare the values of input numbers and produce an output indicating whether the numbers are equal or specifies which of the numbers is greater.





Types of Comparators

Depending on the number of bits, the following are some main types of comparators used in digital circuits –

- 1-Bit Magnitude Comparator
- 2-Bit Magnitude Comparator
- 4-Bit Magnitude Comparator

In 2-bit
A₁ A₀ I₁

Let us discuss each type of comparator in detail.

1-Bit Magnitude Comparator

A **1-bit magnitude comparator** is a logic circuit which can compare two binary numbers of one bit each. It produces an output that indicates the relationship between the two input numbers.

In other words, a 1-bit magnitude comparator is one that compares two 1-bit binary numbers and generates an output showing whether one number is equal to or greater than or less than the other.



Topic : 1-bit magnitude Comparator



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The block diagram of a 1-bit magnitude comparator is shown in the following figure –



1 Bit equal condition
which logic gate
give result

For non
equal
which logic
gate is
used

Here, A and B are the 1-bit input numbers, and L, E, and G are the output lines indicating less than or equal to or greater than relationship between A and B respectively.

Let us understand the working this type of comparator.

If A = 0 and B = 0 or if A = 1 and B = 1, then A = B. It indicates that the two binary numbers are equal. Therefore,

$$E = \bar{A} \cdot \bar{B} + A \cdot B = A \odot B$$

If A = 0 and B = 1, then A < B. This indicates that the binary number A is less than the binary number B. Therefore,

$$L = \bar{A} B$$

If A = 1 and B = 0, then A > B. It indicates that the binary number A is greater than the binary number B. Therefore,

$$G = A \bar{B}$$

XNOR

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

$$Y = \bar{A} \bar{B} + A \cdot B$$



Topic : 1-bit magnitude Comparator

Inputs		Outputs		
A	B	L (A < B)	E (A = B)	G (A > B)
0	0	0	1	0
0	1	1	0	0
1	0	0	0	1
1	1	0	1	0

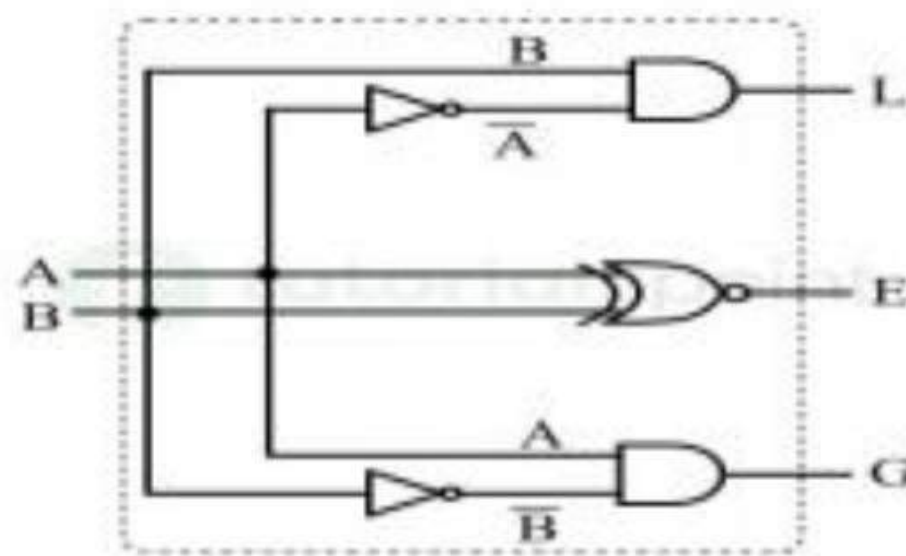
We can use this truth table to obtain the Boolean expression of the 1-bit magnitude comparator.

$$L = \bar{A} B$$

$$E = \bar{A} \cdot \bar{B} + A \cdot B = A \odot B$$

$$G = A \bar{B}$$

The logic circuit diagram of the 1-bit magnitude comparator is shown in the following figure.



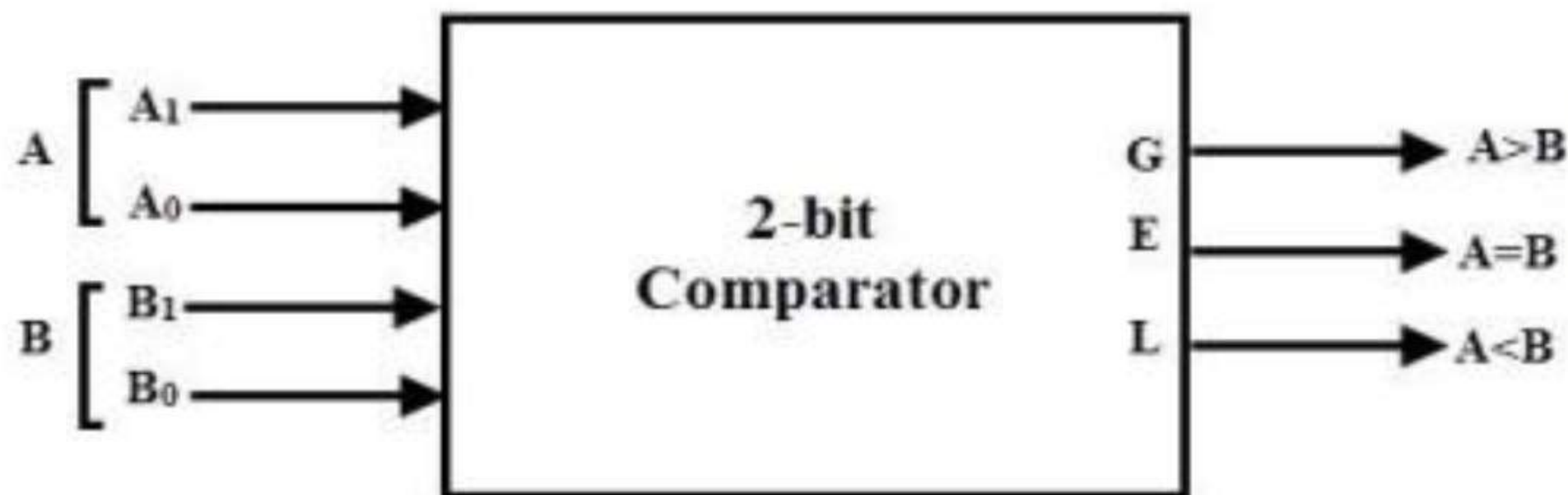


2-Bit Magnitude Comparator

A digital combinational circuit used to compare the magnitudes of two 2-bit binary numbers and determine the relationship between them is called a **2-bit magnitude comparator**.

Hence, the 2-bit magnitude comparator compares the values represented by two 2-bit binary numbers and then generates an output that indicates whether one number is equal to or greater than or less than the other.

The block diagram of a typical 2-bit magnitude comparator is shown in the following figure –



Here, the lines A_0A_1 and B_0B_1 represents two 2-bit binary number inputs and the lines L, E, and G represents the less than, equal to, and greater than output lines.

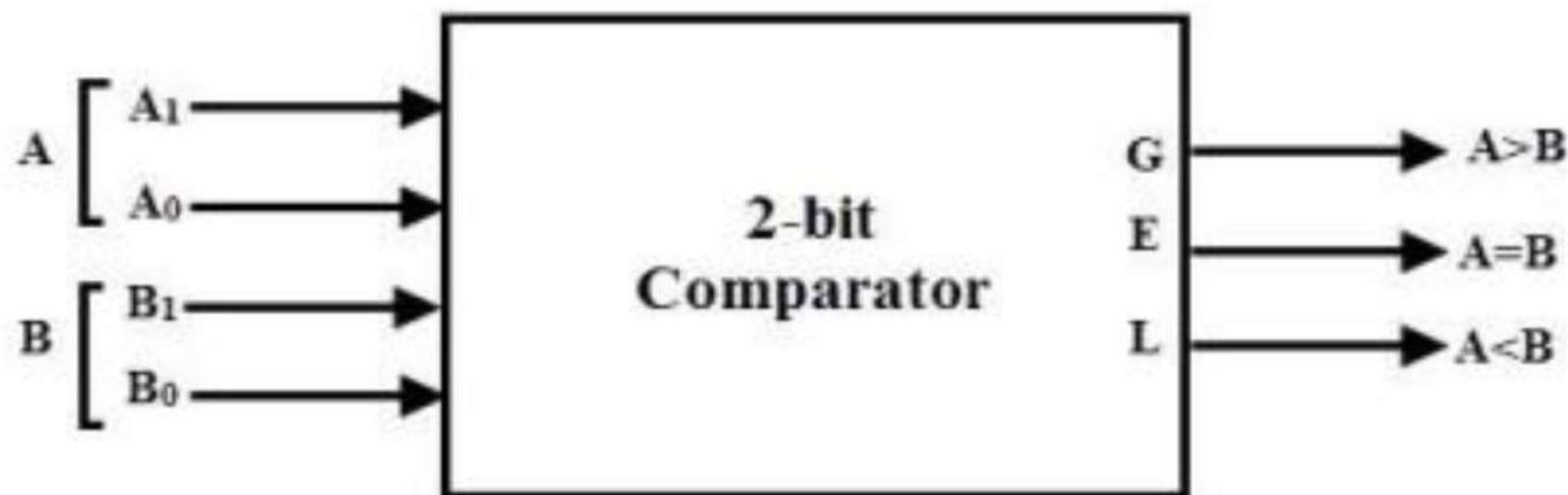


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Here, the lines A_0A_1 and B_0B_1 represents two 2-bit binary number inputs and the lines L, E, and G represents the less than, equal to, and greater than output lines.



Topic : 2-bit magnitude Comparator



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Inputs				Outputs		
A_1	A_0	B_1	B_0	$L (A < B)$	$E (A = B)$	$G (A > B)$
0	0	0	0	0	1	0
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	0	1	0
1	0	1	1	1	0	0
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	0	0	1
1	1	1	1	0	1	0



Topic : 2-bit magnitude Comparator



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Case 1: $A = B$

The comparator produces an output $A = B$ which is E , if $A_0 = B_0$ and $A_1 = B_1$. Therefore, the Boolean expression for the output E will be,

$$E = (A_0 \odot B_0) (A_1 \odot B_1)$$

Case 2: $A < B$

The comparator produces an output $A < B$ which is L , if

- $A_1 = 0$ and $B_1 = 1$, OR
- $A_1 = B_1$ and $A_0 = 0$ and $B_0 = 1$.

From these statements, we can write the Boolean expression for the output L as follows –

$$L = \overline{A_1} B_1 + (A_1 \odot B_1) \overline{A_0} B_0$$

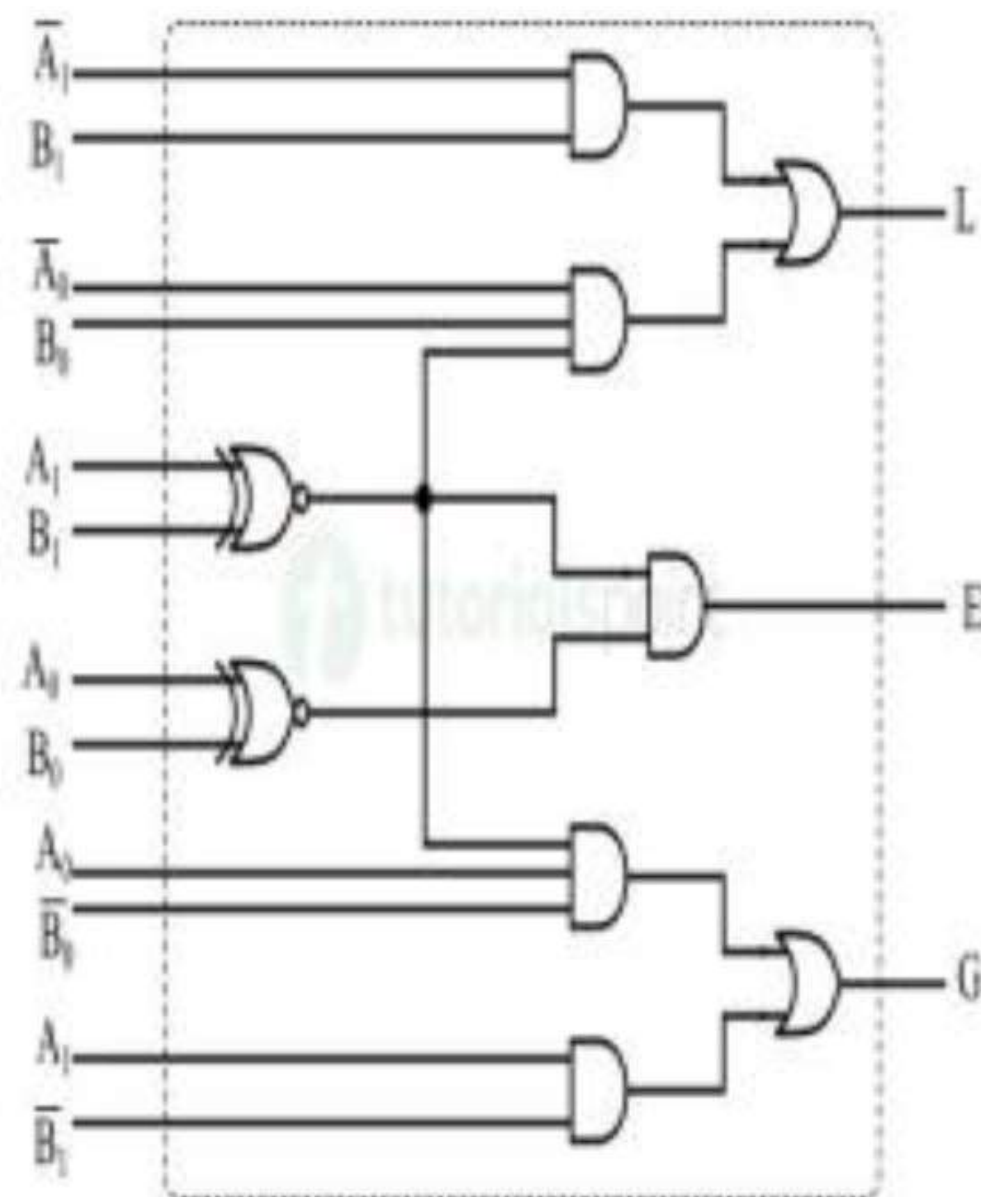
Case 3: $A > B$

The output of the comparator will be $A > B$ i.e., G , if

- $A_1 = 1$ and $B_1 = 0$, OR
- $A_1 = B_1$ and $A_0 = 1$ and $B_0 = 0$.

From these statements, the Boolean expression for the output G will be,

$$G = A_1 \overline{B_1} + (A_1 \odot B_1) A_0 \overline{B_0}$$





Topic :4-bit magnitude Comparator

4-Bit Magnitude Comparator

The 4-bit magnitude comparator is used in more complex digital circuits like microprocessors, microcontrollers, and many more.

It is a type of comparator that can compare the values or magnitudes of two 4-bit binary numbers and produce an output indicating whether one number is equal to or less than or greater than the other.

The block diagram of the 4-bit magnitude comparator is shown in the following figure –



Let us now understand the working of this 4-bit magnitude comparator. For that consider $A = A_3A_2A_1A_0$ is the first 4-bit binary number and $B = B_3B_2B_1B_0$ is the second 4-bit binary number.



Topic :4-bit magnitude Comparator

Case 1: $A = B$

The comparator will produce an output $A = B$ which is E , if all the corresponding bits in the two numbers are equal i.e., $A_3 = B_3$ and $A_2 = B_2$ and $A_1 = B_1$ and $A_0 = B_0$.

In this case, the Boolean expression of the output will be,

$$E = (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) (A_0 \odot B_0)$$

Case 2: $A < B$

The comparator will produce an output $A < B$ which is L , if

- $A_3 = 0$ and $B_3 = 1$, OR
- $A_3 = B_3$ and if $A_2 = 0$ and $B_2 = 1$, OR
- $A_3 = B_3$ and if $A_2 = B_2$ and if $A_1 = 0$ and $B_1 = 1$, OR
- $A_3 = B_3$ and if $A_2 = B_2$ and if $A_1 = B_1$ and if $A_0 = 0$ and $B_0 = 1$.

From these statements, we can derive the Boolean expression for the output L , which is given below.

$$L = \overline{A_3} B_3 + (A_3 \odot B_3) \overline{A_2} B_2 + (A_3 \odot B_3) (A_2 \odot B_2) \overline{A_1} B_1 + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) \overline{A_0} B_0$$



Topic :4-bit magnitude Comparator

Case 1: $A = B$

The comparator will produce an output $A = B$ which is E , if all the corresponding bits in the two numbers are equal i.e., $A_3 = B_3$ and $A_2 = B_2$ and $A_1 = B_1$ and $A_0 = B_0$.

In this case, the Boolean expression of the output will be,

$$E = (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) (A_0 \odot B_0)$$

Case 2: $A < B$

The comparator will produce an output $A < B$ which is L , if

- $A_3 = 0$ and $B_3 = 1$, OR
- $A_3 = B_3$ and if $A_2 = 0$ and $B_2 = 1$, OR
- $A_3 = B_3$ and if $A_2 = B_2$ and if $A_1 = 0$ and $B_1 = 1$, OR
- $A_3 = B_3$ and if $A_2 = B_2$ and if $A_1 = B_1$ and if $A_0 = 0$ and $B_0 = 1$.

From these statements, we can derive the Boolean expression for the output L , which is given below.

$$L = \overline{A_3} B_3 + (A_3 \odot B_3) \overline{A_2} B_2 + (A_3 \odot B_3) (A_2 \odot B_2) \overline{A_1} B_1 + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) \overline{A_0} B_0$$



Topic :4-bit magnitude Comparator



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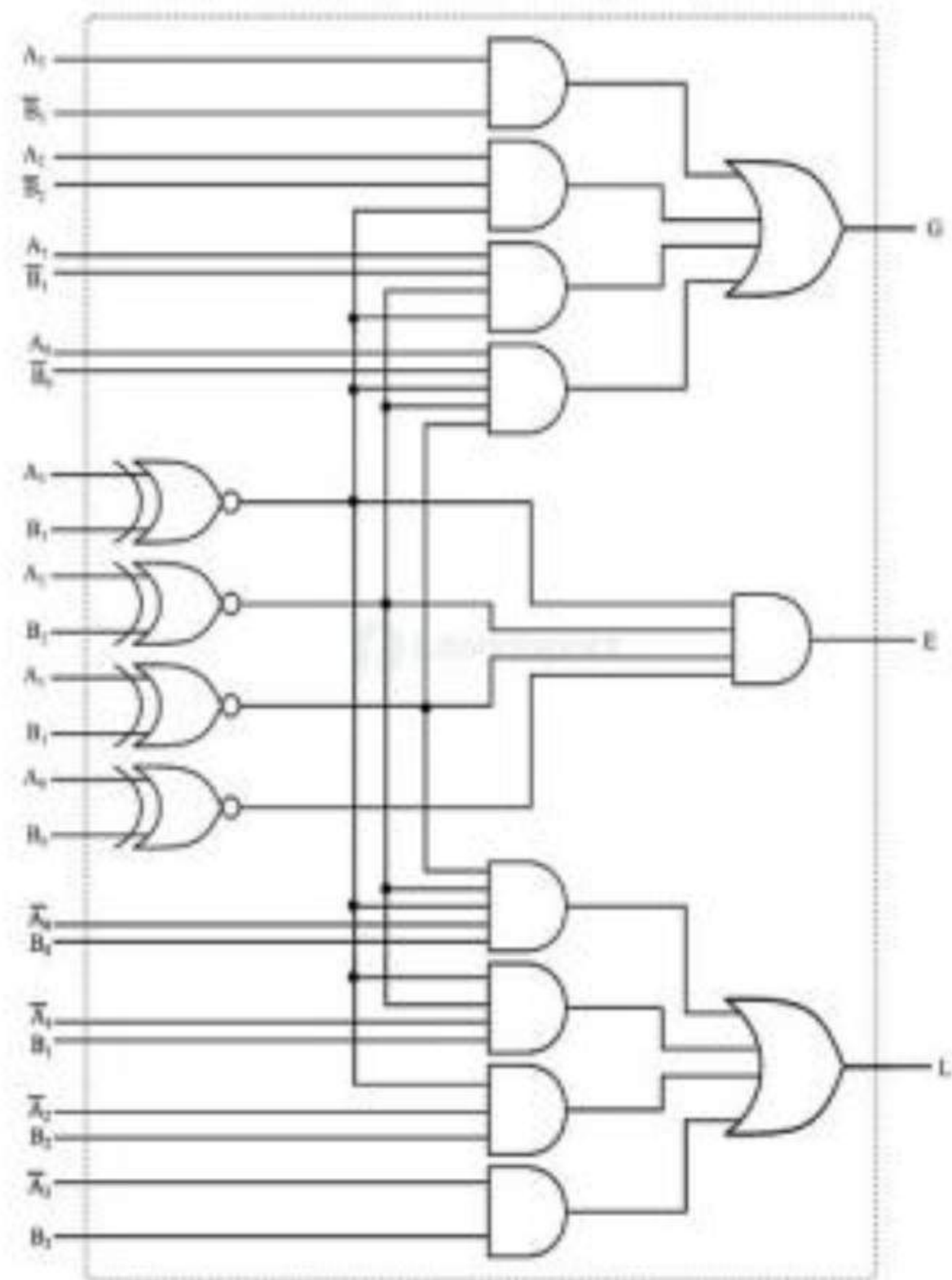
Case 3: $A > B$

The comparator produces an output $A > B$ which is G , if

- $A_3 = 1$ and $B_3 = 0$, OR
- $A_3 = B_3$ and if $A_2 = 1$ and $B_2 = 0$, OR
- $A_3 = B_3$ and if $A_2 = B_2$ and if $A_1 = 1$ and $B_1 = 0$, OR
- $A_3 = B_3$ and if $A_2 = B_2$ and if $A_1 = B_1$ and if $A_0 = 1$ and $B_0 = 0$.

Hence, from these statements, we can write the Boolean expression for the output G which is,

$$G = A_3 \bar{B}_3 + (A_3 \odot B_3) A_2 \bar{B}_2 + (A_3 \odot B_3) (A_2 \odot B_2) A_1 \bar{B}_1 + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) A_0 \bar{B}_0$$





Topic :4-bit magnitude Comparator



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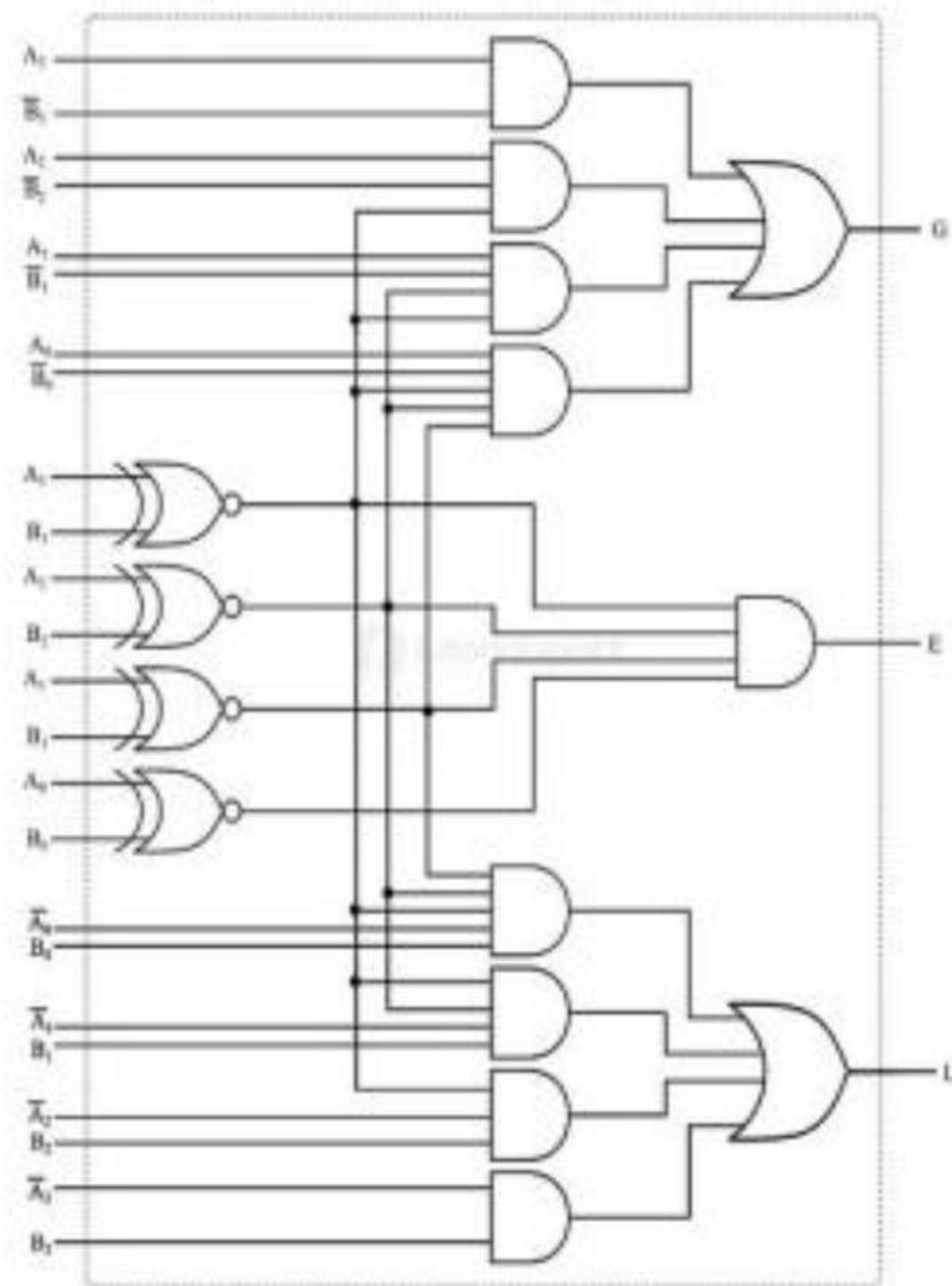
Case 3: $A > B$

The comparator produces an output $A > B$ which is G , if

- $A_3 = 1$ and $B_3 = 0$, OR
- $A_3 = B_3$ and if $A_2 = 1$ and $B_2 = 0$, OR
- $A_3 = B_3$ and if $A_2 = B_2$ and if $A_1 = 1$ and $B_1 = 0$, OR
- $A_3 = B_3$ and if $A_2 = B_2$ and if $A_1 = B_1$ and if $A_0 = 1$ and $B_0 = 0$.

Hence, from these statements, we can write the Boolean expression for the output G which is,

$$G = A_3 \bar{B}_3 + (A_3 \odot B_3) A_2 \bar{B}_2 + (A_3 \odot B_3) (A_2 \odot B_2) A_1 \bar{B}_1 + (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) A_0 \bar{B}_0$$





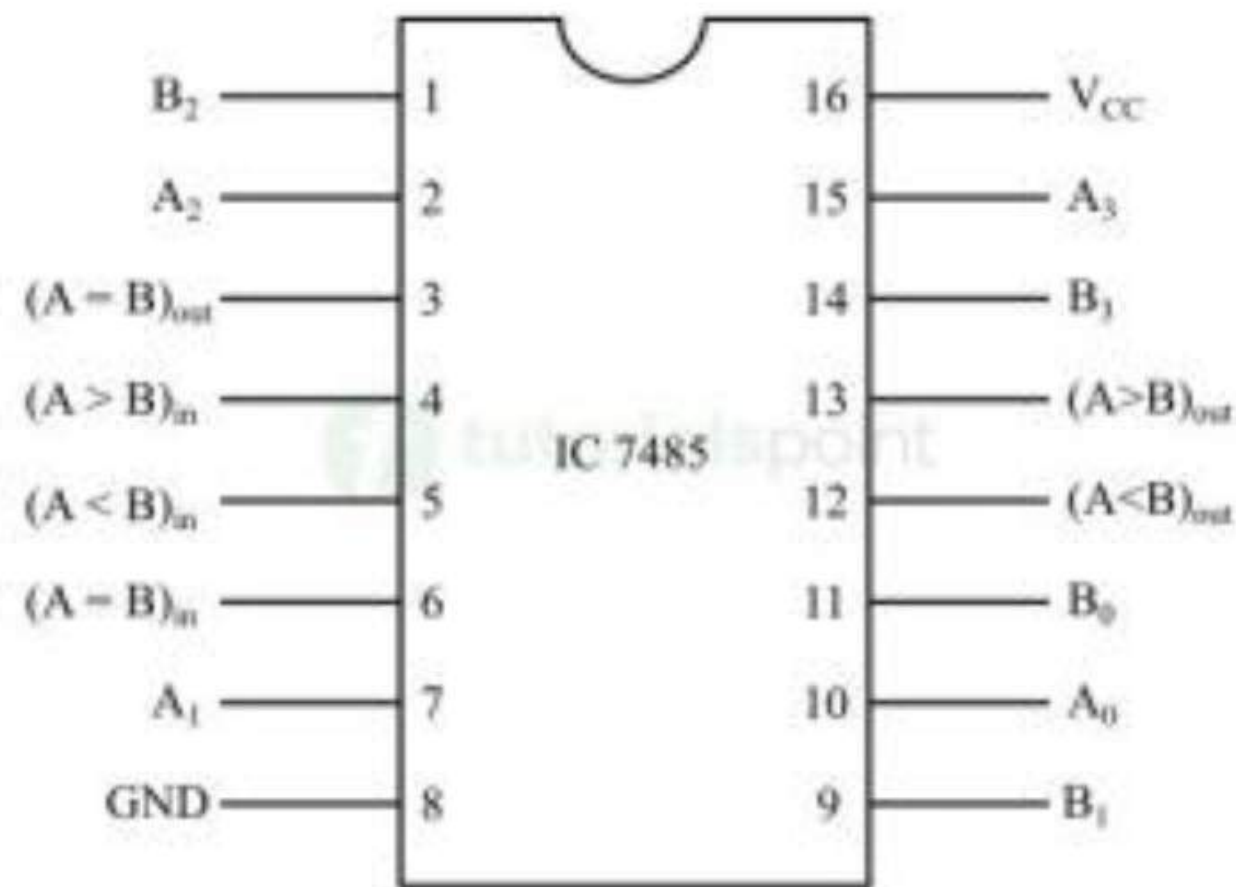
Topic :4-bit magnitude Comparator IC

Comparator IC

A comparator IC is an integrated circuit which is designed to compare two binary numbers and produce an output based on the comparison results.

A commonly used comparator IC is IC 7485 which is a 4-bit magnitude comparator IC. It is widely in digital electronic applications to compare two 4-bit binary numbers.

The IC 7485 compares corresponding bits of the two input numbers and determines whether one number is equal to or greater than or less than the other. The pin diagram of the IC 7485 is depicted in the following figure.



This IC is widely used in various digital circuits such as microprocessors, microcontrollers, control systems, and arithmetic logic units.



Topic: Application and Conclusion

Applications of Comparators

The comparators are fundamental components in various digital circuits. They provide capabilities to compare voltage levels and make decisions.

Some of the key applications of comparators in the field of digital electronics are listed below –

- Comparators are used to detect changes in voltage levels in electronic circuits.
- Comparators are also used temperature monitoring systems.
- Comparators are used as zero-crossing detector in various power circuits like phase control circuits, motor and power control circuits, etc.
- Comparators are key components in analog to digital converters (ADCs).
- In signal conditioning circuits, the comparators are used for amplification and filtering of signals before processing.

Conclusion

In conclusion, a comparator is a combinational logic circuit used in various digital electronic applications to compare two similar quantities like two 2-bit binary numbers or two voltage levels, etc.

Comparators help us to make decisions based on a comparison of two input values. They are widely used in several different electronic devices and systems such as arithmetic logic units, microprocessor-based systems, control systems, automation systems, and telecommunication systems.

Depending on the number of input bits required for a particular application, we can design a 1-bit, 2-bit, or 4-bit comparator. Also, comparators are available in the form of integrated circuit. The most commonly used IC comparator is IC 7485 which is a 4-bit magnitude comparator.

THANK YOU